

Homework 1

via Gradescope

- Failure to submit homework correctly will result in zeroes. *If you do not select the pages of your work correctly, student graders cannot see your work.*
- Handwritten homework is recommended. You do not have to type up your work.
- Problems assigned from the textbook are from the 5th edition.
- No late homework accepted. Lateness due to technical issues will not be excused.

Theorem *. *Let p and q be statement variables. $p \rightarrow q \equiv \neg p \vee q$.*

1. (1 point) Show the logical equivalences using truth tables and say a few words explaining why your truth table shows $\equiv$
 - (a) $\neg(p \vee q) \equiv \neg p \wedge \neg q$ (DeMorgan's law).
 - (b) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ (Distributive law).

Note that your truth table will have 8 rows.
2. (1 point) Prove that $(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$ using Theorem * and Theorem 2.1.1. Annotate your proof. For reference, see example 2.1.14.
3. (1 point) Find all values of p and q for which $p \rightarrow q$ is not equal to $q \rightarrow p$. For which values of p, q are the statement forms equal?
4. (1 point) Show that $[(p \rightarrow q) \wedge (p \rightarrow \neg q)] \rightarrow \neg p$ is a tautology using Theorem * and Theorem 2.1.1. Annotate your proof. For reference, see example 2.1.14.
5. (1 point) Section 2.1 #31.
6. (1 point) Section 2.1 #46(a).
7. (1 point) Section 2.2 #22, 23.
Remark. For both problems, parts (b), (c), (e), (g) only.
8. (1 point) Section 2.2 #14, 38, 43.
Remark. When doing #14, do not use truth tables. Use Theorem * and Theorem 2.1.1.
9. (1 point) Section 2.3 #9, 12(b), 23.

Remark. In addition to the instructions associated to these problems, complete all truth tables even if the argument is invalid. Identify all critical rows even if the argument is invalid.

10. (1 point) Section 2.3 #29, 38(*d*).
11. (1 point) Section 2.3 #42, 44. Annotate.